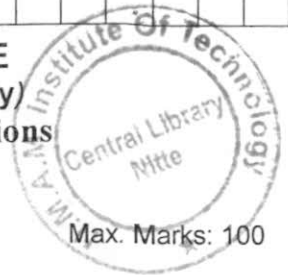


NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023
EC1001-1 – BASIC ELECTRONICS



Duration: 3 Hours

Note:

*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*

*Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. When the p-n junction diode is forward biased, it offers
 - A) low resistance
 - B) high resistance
 - C) low voltage
 - D) high voltage
2. Zener diode is mainly used as
 - A) Comparator
 - B) Voltage regulator
 - C) Oscillator
 - D) Amplifier
3. The efficiency of full wave rectifier is about
 - A) 46 %
 - B) 81.2%
 - C) 1.21%
 - D) 40.6 %
4. What is the collector current for a C-E configuration with a beta of 100 and a base current of 30 μ A?
 - A) 3 mA
 - B) 3 μ A
 - C) 30 Ma
 - D) 30 mA
5. In a CE amplifier circuit, the output signal generates a phase shift of
 - A) 90°
 - B) 180°
 - C) 0°
 - D) 270°
6. For a BJT, the following statement is true
 - A) $I_B = I_C$
 - B) $I_B = I_E$
 - C) $I_E = I_C$
 - D) $I_B = \alpha I_E$
7. Which transistor element is used in CMOS logic?
 - A) MOSFET
 - B) JFET
 - C) Bipolar
 - D) Unijunction
8. Calculate I_D in n- JFET, Given $V_{GS} = -2V$, $V_P = -4V$ and $I_{DSS} = 4mA$
 - A) 2mA
 - B) 3mA
 - C) 4mA
 - D) 1mA
9. Find the output voltage of an ideal op-amp if V_1 and V_2 are the two input voltages
 - A) $V_0 = A(V_1 - V_2)$
 - B) $V_0 = V_1 - V_2$
 - C) $V_0 = A(V_1 + V_2)$
 - D) $V_0 = V_1 \times V_2$
10. A differentiator circuit using an Op Amp hasin its feedback path
 - A) Capacitor
 - B) Inductor
 - C) Diode
 - D) Resistor
11. The voltage levels fixed at one of the terminals of two comparators in the internal circuitry of IC 555 timer are
 - A) $\frac{1}{3}V_{CC}$ & $\frac{2}{3}V_{CC}$
 - B) V_{CC} & $\frac{2}{5}V_{CC}$
 - C) $\frac{1}{5}V_{CC}$ & $\frac{2}{5}V_{CC}$
 - D) $0V$ & $\frac{2}{5}V_{CC}$
12. A sine wave of 0.5 V peak voltage is applied as an input to an inverting amplifier with $R_1 = 10k\Omega$ and $R_f = 50 k\Omega$. The output voltage V_o is
 - A) 5.2 V peak
 - B) -2.5 V peak
 - C) -10 V peak
 - D) -2.5 V peak-peak
13. In Hartley oscillator, the components used in the feedback network are
 - A) 2L and 2C
 - B) 2R and 2C
 - C) 1L and 2C
 - D) 2L and 1C

14. For an amplifier with negative feedback, the closed loop gain is given by
 A) $A_f = \frac{A}{1+A\beta}$ B) $A_f = \frac{A}{1-A\beta}$
 C) $A_f = \frac{A}{1-\beta}$ D) $A_f = \frac{A}{1-V\beta}$
15. In a practical oscillator circuit, to start oscillations, the loop gain $A\beta$ must be
 A) Equal to 1 B) Greater than 1
 C) Less than 1 D) Not equal to 1
16. An amplifier has an open loop voltage gain of 2000. If 40% negative voltage series feedback is used, then the closed loop gain is
 A) 2.49 B) 24.9
 C) 0.249 D) 249
17. Actuators are used to
 A) Sense an object B) Activate a chemical
 C) Make a mechanical movement D) All of the above
18. The function of a sensor is to _____
 A) Detect events within specified environment B) Separate physical parameters
 C) Track and transfer data to computer processor D) Both a and c
19. In a communication system, the noise can be generated at the
 A) transmitter, channel, receiver B) channel, receiver
 C) transmitter, channel D) only in the channel
20. Which of the following is true about optocouplers?
 A) Optocoupler acts as an input device only B) Optocoupler acts as an output device only
 C) Optocoupler can be used in both input and output circuitry D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) For the NPN silicon transistor connected in common emitter configuration, sketch and explain input and output characteristics. | 6 | L2 | 2 | 1 |
| b) Find α_{DC} , β_{DC} and I_B for a transistor with $I_C = 2.5$ mA and $I_E = 2.55$ mA. | 6 | L3 | 2 | 1 |
| c) Explain Zener diode as a voltage regulator with neat circuit diagrams. | 4 | L2 | 1 | 1 |
| 2. a) With reference to a Half Wave Rectifier:
(i) Draw the circuit diagram
(ii) Sketch the waveforms for input ac voltage, load current, load voltage
(iii) Derive the expression for average DC load current and average DC load voltage | 6 | L2 | 1 | 1 |
| b) Explain the operation of a single stage RC coupled amplifier with neat diagrams. Explain significance of each component. | 6 | L2 | 2 | 1 |
| c) Determine the value of R_s in the circuit shown in the Figure Q2(c) when the voltage across the Zener is 12 V and the Zener current is 25 mA. | 4 | L3 | 1 | 1 |

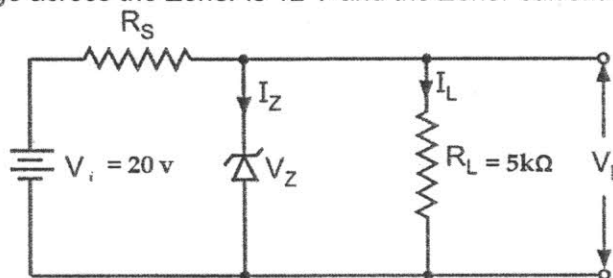


Figure Q2(c)

3. a) Define ripple factor and derive the expression for ripple factor for a half wave rectifier.
- b) For the NPN silicon transistor connected in common base configuration, sketch and explain input and output characteristics.
- c) Derive the equation for the collector current of a PNP transistor in terms of the base current and α_{DC} .

Unit – II

- | | | | | | | |
|----|----|--|---|----|---|---|
| 4. | a) | An Astable Multivibrator using IC555 times generates a square wave of 2 kHz and duty cycle of 75%. Calculate the value of R_A and R_B required. Assume the capacitor of 0.1 μ F is used. | 6 | L3 | 3 | 1 |
| | b) | With neat circuit diagram and necessary equations explain the Hartley oscillator. | 6 | L2 | 4 | 1 |
| | c) | Design a non-inverting amplifier using OP-AMP with a closed loop gain of 10. Assume $R_1 = 1$ k Ω . Also draw the neat circuit diagram. | 4 | L3 | 3 | 1 |
| 5. | a) | Explain the voltage series negative feedback with neat diagram and derive the expression for the closed loop gain. | 6 | L2 | 4 | 1 |
| | b) | Explain the working of a Op-amp inverting amplifier with neat circuit diagram and waveforms. Also derive the expression for closed loop gain. | 6 | L2 | 3 | 1 |
| | c) | Calculate the frequency of oscillations of a Colpitts oscillator with $C_1 = C_2 = 400$ pF and $L = 2$ mH. Also draw the circuit diagram. | 4 | L3 | 4 | 1 |
| 6. | a) | In a Hartley oscillator $L_1 = 2$ mH, $L_2 = 8$ mH, $M = 100$ μ H & $C = 0.001$ μ F. Calculate:
(i) frequency of oscillations
(ii) feedback factor
(iii) voltage gain | 6 | L3 | 4 | 1 |
| | b) | Design an adder circuit using Op-amp to generate an output voltage given by $V_0 = - (1.5 V_1 + 1.8 V_2 + 2.2 V_3)$. Chose $R_f = 10$ k Ω . Also draw the circuit diagram with components. | 6 | L3 | 3 | 1 |
| | c) | State and explain the Barkhausen's criterion for sustained oscillations. | 4 | L2 | 4 | 1 |

Unit – III

- | | | | | | | |
|----|----|--|---|----|---|---|
| 7. | a) | With neat diagram explain the basic elements of an electronic communication system. | 6 | L2 | 5 | 1 |
| | b) | With neat diagrams explain the elements of embedded systems. | 6 | L2 | 5 | 1 |
| | c) | Explain the concept of frequency reuse in cellular communication system. | 4 | L2 | 5 | 1 |
| 8. | a) | What is noise? Discuss briefly about the noise in a communication system with suitable examples. | 6 | L2 | 5 | 1 |
| | b) | Compare the microprocessor and microcontroller. | 6 | L2 | 5 | 1 |
| | c) | If there are 4 groups of channels in a cellular system with a cluster size of 7, then find the total number of duplex channels. If there are 6 clusters covering the geographical area, then find the cellular capacity. | 4 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
